LESSON 11

ESP32 SUPER STARTER KIT · 2-HOUR SESSION

Weather Station

Measure temperature and humidity and show them on your screen.

Name: _____ Date: _____

TODAY YOU WILL LEARN

- How a **sensor** turns the world into numbers
- What **humidity** is
- How to use **float** numbers with a decimal point
- How to make decisions with **if / else if**

SESSION PLAN

0:00	How do sensors work? Install the DHT library
0:15	Wire up the DHT11
0:25	Sketch 1: **L11_ReadDHT** (Serial Monitor)
0:50	Short break
0:55	Sketch 2: **L11_Station** (on the screen)
1:15	Sketch 3: **L11_Comfort** (comfort face)
1:35	Challenges
1:55	Quiz and tidy up

PARTS TO COLLECT

- ☐ ESP32 board + USB-C cable, on your 2 breadboards
- ☐ OLED screen, still wired from Lesson 10
- ☐ DHT11 temperature and humidity module (blue, 3 pins)
- ☐ 3 jumper wires
- ☐ *For a challenge:* thermistor + 10 kΩ resistor (brown-black-orange)

1. Measuring the weather

A **sensor** is a part that measures something in the world and turns it into an electrical signal. Your microcontroller can then read it as a number.

The blue **DHT11** module has two sensors in one:

- a **thermometer** that measures **temperature** in degrees Celsius (°C)
- a **humidity** sensor that measures how much **water vapour** is in the air, in percent (%)

> **Big idea: humidity**
>
> Air always holds some invisible water. 30% humidity feels dry (your lips get chapped). 80% feels damp and sticky, like a bathroom after a hot shower. Most people feel comfortable between about 30% and 60%.

The DHT11 has a tiny chip inside that does the measuring. It sends the answers to the ESP32 as a stream of 1s and 0s on **one wire**. Decoding that stream is tricky, so we use a library.

Install the library:

- ☐ **Tools → Manage Libraries…**
- ☐ Search for **DHT sensor library** (by Adafruit)
- ☐ Click **Install**, then **Install All** to get *Adafruit Unified Sensor* too.

2. Wire up the DHT11

Keep the OLED screen wired up from Lesson 10. Push the DHT11 module into the breadboard and **read the labels** on its pins. They are usually **S** (signal), **+** and **–**.

✓	DHT11 pin	ESP32 pin
☐	**S** (signal / DATA / OUT)	**4**
☐	**+** (VCC)	**3V3**
☐	**–** (GND)	**GND**

> **Check before you plug in**
>
> If + and – are swapped, the sensor gets hot and can be damaged. Double-check, then plug in the USB cable.

3. Sketch 1: Read the sensor

Type this in and save it as **L11_ReadDHT**. Open the **Serial Monitor** (115200 baud).

Type this in → save it as **L11_ReadDHT** — 27 lines

```
// Lesson 11: Read temperature and humidity
// DHT11 sensor: data pin on pin 4.

#include <DHT.h>

DHT dht(4, DHT11);

void setup() {
  Serial.begin(115200);
  dht.begin();
}

void loop() {
  delay(2000);  // the DHT11 needs 2 seconds between readings
  float temp = dht.readTemperature();
  float humidity = dht.readHumidity();

  if (isnan(temp) || isnan(humidity)) {
    Serial.println("Could not read the sensor!");
    return;
  }
  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.print(" C   Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
}
```

Code	What it means
`#include <DHT.h>`	Use the DHT library
`DHT dht(4, DHT11);`	Make a sensor called **dht**, on pin 4, of type **DHT11**
`float temp`	A **float** is a number that can have a decimal point, like 23.5
`dht.readTemperature()`	Ask the sensor for the temperature in °C
`isnan(temp)`	"Is Not A Number?" This is true if the reading failed
`return;`	Stop this turn of `loop()` early, then start again from the top

Try it: be a scientist

1. What temperature and humidity does your sensor show?

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2. **Breathe** gently on the sensor for 5 seconds. What happens to the humidity? Why?

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3. Hold the sensor between your fingers (not on the pins). How high does the temperature get?

Serial Monitor says: Could not read the sensor?

Check the S wire is on pin **4**. Wait a few seconds after uploading, because the sensor is slow to wake up.

4. Sketch 2: Weather station

Now let's show the readings on the screen, so your weather station works without a computer.

No computer needed

Once it's uploaded, your sketch stays in the ESP32. Unplug it from the computer and power it from a USB phone charger. Now it's a real weather station you can put anywhere!

Save as **L11_Station**:

Type this in → save it as **L11_Station** — 37 lines

```
// Lesson 11: Weather station
// DHT11 on pin 4, OLED screen on pins 21 (SDA) and 22 (SCL).

#include <DHT.h>
#include <Adafruit_SSD1306.h>

DHT dht(4, DHT11);
Adafruit_SSD1306 display(128, 64, &Wire, -1);

void setup() {
  dht.begin();
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
```

```
  display.setTextColor(SSD1306_WHITE);
}

void loop() {
  float temp = dht.readTemperature();
  float humidity = dht.readHumidity();

  display.clearDisplay();
  display.setTextSize(1);
  display.setCursor(0, 0);
  display.println("Weather station");
  if (isnan(temp) || isnan(humidity)) {
    display.println("Sensor error!");
  } else {
    display.setTextSize(2);
    display.setCursor(0, 18);
    display.print(temp, 1);
    display.print(" C");
    display.setCursor(0, 42);
    display.print(humidity, 0);
    display.print(" %");
  }
  display.display();
  delay(2000);
}
```

Code	What it means
Two `#include` lines	One sketch can use **two libraries** at once
`display.print(temp, 1);`	Print with **1** number after the decimal point: 23.4
`display.print(humidity, 0);`	Print with **no** decimal places: 45
`if (…) { … } else { … }`	If the reading failed, show an error. **Otherwise** show the numbers

5. Sketch 3: Comfort face

Let's make the weather station tell us **how the room feels**, with a smiley or a sad face.

Save as **L11_Comfort**:

Type this in → save it as **L11_Comfort** — 57 lines

```
// Lesson 11: Comfort face
// A smiley face when the room is comfortable, a sad face when not.

#include <DHT.h>
#include <Adafruit_SSD1306.h>

DHT dht(4, DHT11);
Adafruit_SSD1306 display(128, 64, &Wire, -1);
```

```
void drawFace(bool happy) {
  if (happy) {
    display.fillCircle(100, 38, 10, SSD1306_WHITE);  // smile
    display.fillCircle(100, 34, 10, SSD1306_BLACK);
  } else {
    display.fillCircle(100, 52, 10, SSD1306_WHITE);  // frown
    display.fillCircle(100, 56, 10, SSD1306_BLACK);
  }
  display.drawCircle(100, 36, 22, SSD1306_WHITE);
  display.fillCircle(92, 28, 3, SSD1306_WHITE);
  display.fillCircle(108, 28, 3, SSD1306_WHITE);
}

void setup() {
  dht.begin();
  display.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  display.setTextColor(SSD1306_WHITE);
}

void loop() {
  float temp = dht.readTemperature();
  float humidity = dht.readHumidity();

  String message = "Just right!";
  if (temp > 26) {
    message = "Too hot!";
  } else if (temp < 18) {
    message = "Too cold!";
  } else if (humidity > 60) {
    message = "Too damp!";
  } else if (humidity < 30) {
    message = "Too dry!";
  }

  display.clearDisplay();
  display.setTextSize(2);
  display.setCursor(0, 4);
  display.print(temp, 0);
  display.println(" C");
  display.print(humidity, 0);
  display.println(" %");
  display.setTextSize(1);
  display.setCursor(0, 52);
  display.print(message);
  drawFace(message == "Just right!");
  display.display();
  delay(2000);
}
```

Code	What it means
`String message = "Just right!";`	A **String** variable holds text. We start by guessing "Just right!"
`if … else if … else if …`	Check one thing after another. Only the **first** true one happens
`message == "Just right!"`	`==` compares: is the message *exactly* "Just right!"? This is either true or false
`void drawFace(bool happy)`	A **bool** is just true or false. The function draws a smile for true and a frown for false

Big idea: the black circle trick

There's no "draw a smile" command! So we draw a white circle, then a **black** circle slightly higher on top of it. Only a thin crescent of white is left: a smile. For a frown, the black circle goes slightly **lower**.

Try it

1. Breathe on the sensor until the face turns sad. What message do you get?
2. Change the numbers so that 22 °C counts as "Too hot!". Can you make the face sad by holding the sensor?

6. Challenges

Challenge 1: Fahrenheit

Show the temperature in °F as well. *Hint:* `dht.readTemperature(true)` *gives Fahrenheit.*

Challenge 2: Record breaker

Remember the **lowest** and **highest** temperature since the ESP32 started, and show them on the screen. *Hint: start with* `float lowest = 100;`.

Challenge 3: Warning light

Add a red LED (with a 220 Ω resistor) on pin **26** that switches on when it's "Too hot!".

Bonus: Thermistor thermometer

A **thermistor** is a resistor that changes with temperature. Wire: **3V3 → 10 kΩ resistor → pin 34 → thermistor → GND**. Read it with `analogReadMilliVolts(34)` and watch the number change when you hold it. Ask your teacher for the formula to turn it into °C, and compare it with the DHT11!

7. Quiz

Check what you learned

1. What two things does the DHT11 measure?

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2. What is the difference between an `int` and a `float`?

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3. Why did the humidity go up when you breathed on the sensor?

Tidy up: unplug the USB cable and put the DHT11 back in the kit. You can leave the screen wired for later lessons.